

Kaustav Purkayastha

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🌐 [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SKILLS

Languages & Data: SQL, PL/SQL, Redshift SQL, Shell Scripting, Python, PySpark; ETL/ELT, CDC, SCD Type-1 & Type-2, Pushdown Optimization, Data Warehousing, Data Modeling, Data Quality & Validation

Cloud & Platforms: AWS (Redshift, S3, Glue, EC2, Aurora), Databricks, Snowflake, dbt, Informatica IDMC/IICS, MS SQL Server, PostgreSQL, Oracle

Tools & DevOps: Stonebranch, Git/GitHub, GitHub Actions, Jenkins, Liquibase, Docker, Jira (Agile/Scrum), Apache Airflow, Amazon Bedrock

WORK EXPERIENCE

Deloitte Consulting, Hyderabad Oct 2023 – Present
Data Engineer – AI & Data Practice

- Engineered the Retail Annuities cloud data platform for a Fortune 500 insurer to replace a legacy on-prem reporting warehouse, building **70+ ETL jobs** in **Informatica IDMC** that integrate multiple enterprise source systems (SQL Server, mainframe, and valuation) and load 35+ years of history into **60+ Redshift tables** using **CDC and SCD logic**.
- Designed historical-load logic to integrate **multi-grain history** (yearly, monthly, and point-in-time snapshots) into the **Redshift data model**, performing deep source analysis to derive record effective dates from a complex source of main and archive tables while preserving natural keys and audit integrity.
- Migrated 4+ legacy Tableau Prep workflows into scalable **Redshift ELT pipelines** using **pushdown optimization**, consolidating multiple RDBMS, external, and file (CSV) sources and re-engineering complex Python annuity calculations into optimized SQL, then validating outputs to **~98% accuracy** against legacy business extracts.
- Migrated interdependent **Dremio views to Redshift materialized views** under tight deadlines, analyzing and untangling deep hierarchical dependencies and performing full **data reconciliation** to confirm parity with source.
- Designed a reusable, configurable looping framework in **Informatica IDMC** that ingests variable-count daily mainframe feeds file-by-file, applying **SCD** sequentially within a single run to preserve correct change history when multiple files arrive on the same day, scheduled via **Stonebranch**.
- Applied **Redshift ELT pushdown optimization** to re-engineer compute-intensive IICS ETL processing into **post-SQL** executed directly on Redshift, removing IICS compute and reducing runtime, using Claude via Amazon Bedrock to accelerate the conversion.

PERSONAL PROJECTS

HealthPipe AI [View Source Code](#) Mar 2026 – May 2026
Multi-agent pipeline that ingests **healthcare data from public APIs**, processes it through **four AI agents** powered by local LLMs, and loads clean output into **DuckDB**. Includes **quality scorecards**, auto-generated **data dictionaries**, and **PySpark** support for large-scale datasets.
Python, DuckDB, PySpark, Ollama, Pytest, GitHub Actions

TestNucleus [View Source Code](#) Jun 2025 – Aug 2025
Config-driven **data quality framework** that separates test definitions (**YAML**) from execution logic. Supports **20+ validation checks** across completeness, uniqueness, and referential integrity — built for **CI/CD and pipeline integration**.
Python, SQLAlchemy, Pydantic, Pytest

EDUCATION

Lovely Professional University 2024 – 2026
Master of Computer Applications, Data Science & Finance CGPA: 7.48

Assam University 2020 – 2023
Bachelor of Science with Honours, Computer Science CGPA: 8.86

CERTIFICATIONS & RECOGNITION

- Certifications:** [Snowflake SnowPro Core](#) (2026), [Databricks Certified Data Engineer Associate](#) (2025), [Claude with Amazon Bedrock](#) (2026), [Industry Proficiency: Financial Services](#) (2025), [IDMC Associate](#) (2024)
- Recognition:** Outstanding Performance Award (Feb 2026, May 2025) | Applause Award (Jul 2024) | FY24 Rising Stars (May 2024) | Institutional Merit Scholarship (Dec 2024)